

MCQs

Set 'A'

Group 'A'

By Pankaj Purbey

1. (A) 8

2. (B) $2^{2500} [1 \times 3 \times 5 \times \dots \times 4999]$

3. (D) 0°

4. (C) $2x - 3y = 5 ; x + 6y = 5$

5. (A) $\perp$ unit

6. (B) $y = 3$

7. (C) 3

8. (D)
$$\left(\begin{array}{ccc|c} 3 & 0 & 2 & 1 \\ 5 & 3 & 0 & 1 \\ -2 & 6 & 0 & -5 \end{array} \right)$$

9. (B) $\frac{1}{\sqrt{2}} \sin^{-1} \sqrt{\frac{2}{3}} x + C$

10. (D) None of these

11. (A) 2, 5

MCQs

Set 'B'

Group 'A'

1. (D) Both A and B

2. (B) $2^{250} [1 \times 3 \times 5 \times \dots \times 499]$ or, (C) $2^{250} [1 \times 3 \times 5 \times \dots \times 499]$

3. (C) 270°

4. (A) $2x - 3y = 5 ; 4x - 6y = 10$

5. (C) $y = 0$

6. (A) $\frac{1}{\sqrt{2}} \sin^{-1} \sqrt{\frac{2}{3}} x + C$

7. (D) None of these

8. (C) 2, 3

9. (D) $\perp$ unit

10. (C) 3

11. (B)
$$\left(\begin{array}{ccc|c} 3 & 0 & -2 & -1 \\ 5 & 3 & 0 & 1 \\ 2 & 6 & 0 & -5 \end{array} \right)$$

Group 'B'

Q.N. 12. (a) Ans: Total number of distinct permutations $\rightarrow$

$$\frac{n!}{r!}$$

12. (b) Ans: The relationship between $p(n, n)$ and $p(n, n-1) \rightarrow$

$$p(n, n) = p(n, n-1)$$

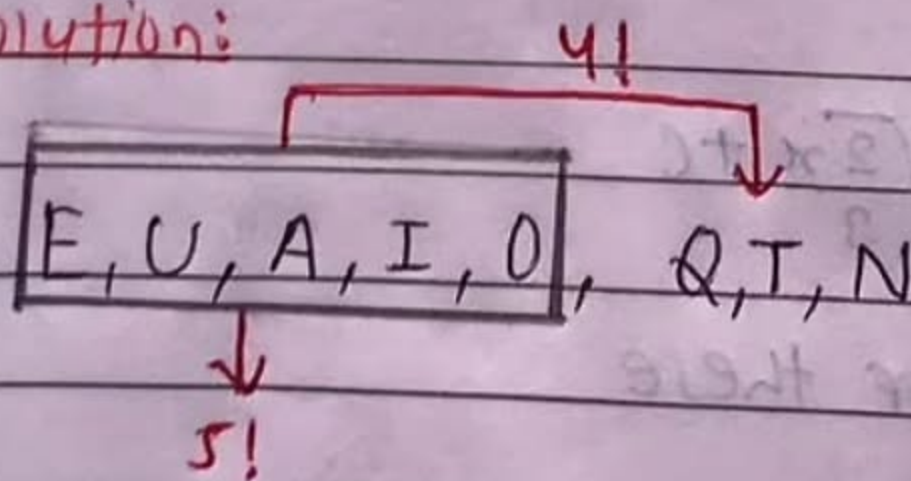
12. (c) Ans: Addition principle of counting states that, "if an event E_1 can occur in m_1 ways, E_2 can occur in m_2 ways, and so on either of these events $E_1, E_2, \dots, E_k$ can occur in $m_1 + m_2 + m_3 + \dots + m_k$ number of ways."

12. (d) Ans: The exponential form of complex number
 $r(\cos \theta^\circ + i \sin \theta^\circ) \rightarrow re^{i \frac{\theta^\circ}{180}}$

12. (e) Ans: The necessary condition for having solution of the system of linear equations by Cramer's Rule,
 $D \neq 0$.

Q.N. 13. (a) solution:

(i)

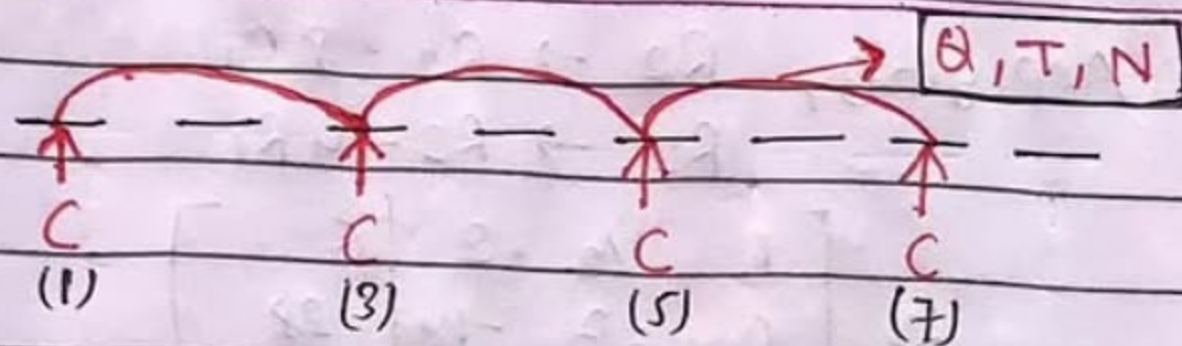


$\therefore$ No. of words that can be formed so that all the vowels are together = $5! \times 4! = 2880$ //

P.T.O. $\rightarrow$

EQUATION

(ii)



∴ No. of words that can be formed so that consonants occupy odd places = $4P_3 \times 5! = 2880 //$

13. (b) Solution: Given digits, 0, 2, 3, 5, 6, 8

Case I: do not end with 0,

$$\frac{4}{3,5} \frac{4}{2,6,8} \frac{3}{2 \text{ remaining}} = 48$$

Case II: end with 0,

$$\frac{5}{2,3,5,6,8} \frac{4}{40} \frac{1}{0} = 20$$

∴ Total no. of even numbers = $48 + 20 = 68 //$

Q.N. 14. Solution:

Given equations,

$$2x + y + 4z = 12$$

$$8x - 3y + 2z = 20$$

$$4x + 11y - z = 33$$

Using forward elimination,
the augmented matrix,

$$\left[\begin{array}{ccc|c} 2 & 1 & 4 & 12 \\ 8 & -3 & 2 & 20 \\ 4 & 11 & -1 & 33 \end{array} \right]$$

Applying, $R_1 \rightarrow R_1/2$

$$\sim \left[\begin{array}{ccc|c} 1 & 1/2 & 2 & 6 \\ 8 & -3 & 2 & 20 \\ 4 & 11 & -1 & 33 \end{array} \right]$$

Applying, $R_2 \rightarrow R_2 - 8R_1$
 $R_3 \rightarrow R_3 - 4R_1$

$$\sim \begin{bmatrix} 1 & 1/2 & 2 & 6 \\ 0 & -7 & -14 & -28 \\ 0 & 9 & -9 & 9 \end{bmatrix}$$

Applying, $R_2 \rightarrow R_2 / -7$

$$\sim \begin{bmatrix} 1 & 1/2 & 2 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 9 & -9 & 9 \end{bmatrix}$$

Applying, $R_3 \rightarrow R_3 - 9R_2$

$$\sim \begin{bmatrix} 1 & 1/2 & 2 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & -27 & -27 \end{bmatrix}$$

Using backward substitution,
Corresponding equations,

$$x + \frac{y}{2} + 2z = 6 \quad \dots (1)$$

$$y + 2z = 4 \quad \dots (2)$$

$$-27z = -27 \quad \dots (3)$$

from eqn (3),

$$-27z = -27$$

$$\therefore z = 1$$

from eqn (2),

$$y + 2z = 4$$

$$\text{or, } y + 2 \times 1 = 4$$

$$\therefore y = 2$$

from eqn (1),

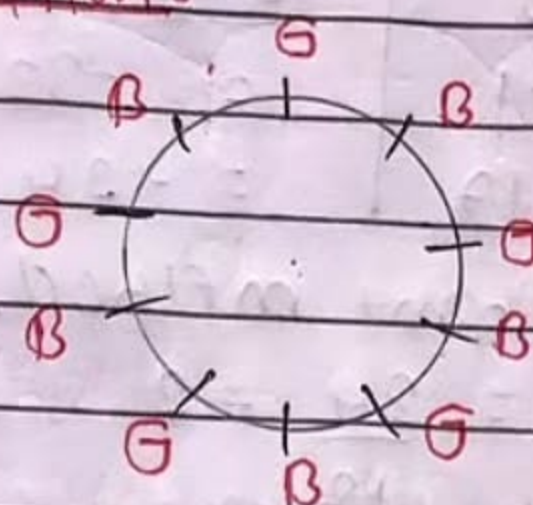
$$x + y/2 + 2z = 6$$

$$\text{or, } x + 2/2 + 2 \times 1 = 6$$

$$\therefore x = 3$$

Hence, $x = 3, y = 2, z = 1$ //

Q.N. 15. (a) solution:



No. of ways boys can sit = $(5-1)! = 4!$

No. of ways girls can sit = $5!$

$\therefore$ Total no. of ways = $4! \times 5! = 2880 //$

15. (b) solution: Given equations,

$$2x + y - z = 9$$

$$3x - y + 2z = -1$$

$$4x + \cancel{y} - 3z = 17$$

Coeff. of x	Coeff. of y	Coeff. of z	Constants
2	1	-1	9
3	-1	2	-1
4	1	-3	17

Now,

$$D = \begin{vmatrix} 2 & 1 & -1 \\ 3 & -1 & 2 \\ 4 & 1 & -3 \end{vmatrix} = 2(3-2) - 1(-9-8) - 1(3+4) \\ = 2 + 17 - 7 \\ = 12$$

$$D_x = \begin{vmatrix} 9 & 1 & -1 \\ -1 & -1 & 2 \\ 17 & 1 & -3 \end{vmatrix} = 9(3-2) - 1(3-34) - 1(-1+17) \\ = 9 + 31 - 16 \\ = 24$$

$$D_y = \begin{vmatrix} 2 & 9 & -1 \\ 3 & -1 & 2 \\ 4 & 17 & -3 \end{vmatrix} = 2(3-34) - 9(-9-8) - 1(51+4) \\ = -62 + 153 - 55 \\ = 36$$

$$\Delta z = \begin{vmatrix} 2 & 1 & 9 \\ 3 & -1 & -1 \\ 4 & 1 & 17 \end{vmatrix} = 2(-17+1) - 1(51+4) + 9(3+4)$$

$$= -32 - 55 + 63$$

$$= -24$$

Using Determinant method,

$$x = \frac{\Delta x}{\Delta} = \frac{24}{12} = 2$$

$$y = \frac{\Delta y}{\Delta} = \frac{36}{12} = 3$$

$$z = \frac{\Delta z}{\Delta} = \frac{-24}{12} = -2$$

Hence, $x=2$, $y=3$, $z=-2$. //

Q.N. 16. (a) Ans: $\int \operatorname{sech} x dx = \ln \left| \tanh \frac{x}{2} \right| + c$

16. (b) Ans: The relation between $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)}$ and $\lim_{x \rightarrow 1} \frac{f'(x)}{g'(x)}$

$$\lim_{x \rightarrow 1} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 1} \frac{f'(x)}{g'(x)}$$

16. (c) Ans: IF $\frac{d}{dx} (F(x) + c) = f(x)$ then,

$$\int f(x) dx = F(x) + c$$

16. (d) Ans: The equation of tangent to curve $y = g(x)$ at (h, k) which is parallel to y -axis, $\rightarrow x = h$. //

16. (e) Ans: $\frac{d \tanh^{-1} u}{du} = \frac{1}{1-u^2}$, $|u| < 1$

P.T.O. $\rightarrow$

Q. 14.13. (a) solution:

Since n is a natural number and multiple of 3.

$$n = 3k, \quad k = 1, 2, 3, \dots$$

Also, $\frac{-1 + i\sqrt{3}}{2} = \omega$ is cube root of unity.

$$\text{L.H.S.} = 1 + \left(\frac{-1 + i\sqrt{3}}{2} \right)^n + \left(\frac{-1 - i\sqrt{3}}{2} \right)^n$$

$$= 1 + \omega^n + \omega^{2n}$$

$$= 1 + \omega^{3k} + \omega^{2 \times 3k}$$

$$= 1 + (\omega^3)^k + (\omega^3)^{2k}$$

$$= 1 + 1^k + 1^{2k}$$

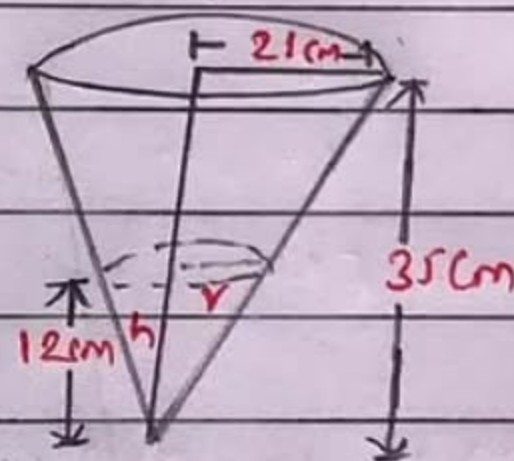
$$= 1 + 1 + 1$$

$$= 3 \Rightarrow \text{R.H.S.} \quad //$$

17. (b) solution:

Let ' v ' be volume of water, ' r ' be radius of base and ' h ' be height of tank at time t .

$$\frac{dv}{dt} = 36 \text{ cm}^3/\text{min}$$



$$\text{Also, } \frac{35}{h} = \frac{21}{r}$$

$$\therefore r = \frac{21}{35} h = \frac{3}{5} h$$

we have,

$$\text{Volume } (V) = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi \left(\frac{3}{5} h \right)^2 h = \frac{1}{3} \pi \times \frac{9}{25} h^3$$

Differentiating both sides w.r.t ' t ',

$$\frac{dv}{dt} = \frac{1}{3} \pi \times \frac{9}{25} \frac{dh^3}{dt}$$

$$\text{or, } \frac{dv}{dt} = \frac{1}{3} \pi \times \frac{9}{25} \times 3h^2 \times \frac{dh}{dt}$$

or, $36 = \frac{1}{3} \pi \times \frac{9}{25} \times 3 \times (12)^2 \frac{dh}{dt}$

or, $\frac{dh}{dt} = \frac{36 \times 25}{1296\pi}$

$\therefore \frac{dh}{dt} = \frac{25}{36\pi} \text{ cm/min} //$

Q.N. 18. (a) solution: Given,

Radius (r) = 7 cm

$\Delta r = dr = 7.1 - 7 = 0.1 \text{ cm}$

We have,

Area of circle (A) = πr^2

$dA = 2\pi r dr$

$\therefore dA = 2\pi \times 7 \times 0.1 = \frac{7\pi}{5} \text{ cm}^2 //$

18. (b) solution:

$\int \frac{t^2}{t^4 - 2t^2 - 15} dt$

Let, $t^2 = a$. Then,

$\frac{t^2}{t^4 - 2t^2 - 15} = \frac{a}{a^2 - 2a - 15} = \frac{a}{(a-5)(a+3)}$

Let, $\frac{a}{(a-5)(a+3)} = \frac{A}{(a-5)} + \frac{B}{(a+3)}$

or, $a = A(a+3) + B(a-5)$

put $a = -3$,

$-3 = A \times 0 + B \times -8$

$\therefore B = \frac{3}{8}$

Also, put $a = 5$,

$5 = A \times 8 + B \times 0$

$\therefore A = \frac{5}{8}$

Date _____
Page _____

$$\int \frac{t^2}{t^4 - 2t^2 - 15} dt = \frac{a}{(a-5)(a+3)} = \frac{A}{a-5} + \frac{B}{a+3}$$

$$= \int \left(\frac{5}{8} \frac{1}{t^2-5} + \frac{3}{8} \frac{1}{t^2+3} \right) dt$$

$$= \frac{5}{8} \int \frac{1}{t^2-5} dt + \frac{3}{8} \int \frac{1}{t^2+3} dt$$

$$= \frac{5}{8} \times \frac{1}{2\sqrt{5}} \ln \left| \frac{t-\sqrt{5}}{t+\sqrt{5}} \right| + \frac{3}{8} \times \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{t}{\sqrt{3}} \right) + C$$

$$= \frac{\sqrt{5}}{16} \ln \left| \frac{t-\sqrt{5}}{t+\sqrt{5}} \right| + \frac{\sqrt{3}}{8} \tan^{-1} \left(\frac{t}{\sqrt{3}} \right) + C //$$

Q.N. 19. (a) solution:

$$y = e^{\sinh^{-1}x}$$

Differentiating on both sides w.r.t. 'x',

$$\frac{dy}{dx} = \frac{d e^{\sinh^{-1}x}}{dx}$$

$$\text{or, } \frac{dy}{dx} = \frac{d e^{\sinh^{-1}x}}{d \sinh^{-1}x} \cdot \frac{d \sinh^{-1}x}{dx}$$

$$\therefore \frac{dy}{dx} = e^{\sinh^{-1}x} \cdot \frac{1}{\sqrt{1+x^2}} //$$

19. (b) solution: Given,

$$\sqrt{1-x^2} dy + \sqrt{1-y^2} dx = 0$$

$$\text{or, } \sqrt{1-x^2} dy = -\sqrt{1-y^2} dx$$

$$\text{or, } \frac{-1}{\sqrt{1-y^2}} dy = \frac{1}{\sqrt{1-x^2}} dx$$

$$\text{or, } -\sin^{-1}y = \sin^{-1}x - \sin C$$

$$\text{or, } \sin^{-1}x + \sin^{-1}y = \sin C$$

$$\text{or, } \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2}) = \sin C$$

$$\therefore x\sqrt{1-y^2} + y\sqrt{1-x^2} = C \quad \text{proved} //$$

Group 'C'

Q.N.20. (a) solution:

$$\frac{\left(\frac{\sin \pi}{8} - i \frac{\cos \pi}{8}\right)^8}{\left(\frac{\sin \pi}{8} + i \frac{\cos \pi}{8}\right)^8}$$

$$= \left[\frac{i \frac{\sin \pi}{8} - i^2 \frac{\cos \pi}{8}}{i \frac{\sin \pi}{8} + i^2 \frac{\cos \pi}{8}} \right]^8$$

$$= \left[\frac{\frac{\cos \pi}{8} + i \frac{\sin \pi}{8}}{-\frac{\cos \pi}{8} + i \frac{\sin \pi}{8}} \right]^8$$

$$= \left(\frac{\cos \pi + i \sin \pi}{8} \right)^8$$

$$\frac{\left[- \left(\frac{\cos \pi}{8} - i \frac{\sin \pi}{8} \right) \right]^8}{\left[\left(\frac{\cos \pi}{8} - i \frac{\sin \pi}{8} \right) \right]^8}$$

$$= \frac{\cos \pi + i \sin \pi}{\left[\left(\cos \left(-\frac{\pi}{8} \right) + i \sin \left(-\frac{\pi}{8} \right) \right) \right]^8}$$

$$= \frac{\cos \pi + i \sin \pi}{(\cos(-\pi) + i \sin(-\pi))}$$

$$= \frac{-1 + 0}{-1 - 0} = -1$$

$$= 1 //$$

P.T.O. →

20. (b) solution: $z^3 = 8i$

$$z = (8i)^{1/3}$$

$$1 + 8i = x + iy = r(\cos \theta + i \sin \theta)$$

$$\therefore x = 0, y = 8$$

$$\tan \theta = \frac{y}{x} = 8/0 = 90^\circ$$

$$r = \sqrt{0^2 + 8^2} = 8$$

Using De-Moivre's theorem,

$$z = r^{1/n} \left[\cos \left(\frac{2\pi k + \theta}{n} \right) + i \sin \left(\frac{2\pi k + \theta}{n} \right) \right]$$

$$= 8^{1/3} \left[\cos \left(\frac{2\pi k + 90^\circ}{3} \right) + i \sin \left(\frac{2\pi k + 90^\circ}{3} \right) \right]$$

When $k=0$,

$$z_1 = 8^{1/3} \left[\cos \left(\frac{2\pi \times 0 + 90^\circ}{3} \right) + i \sin \left(\frac{2\pi \times 0 + 90^\circ}{3} \right) \right]$$

$$= 2 \left[\frac{\sqrt{3}}{2} + i \frac{1}{2} \right]$$

$$= \sqrt{3} + i$$

When $k=1$,

$$z_2 = 8^{1/3} \left[\cos \left(\frac{2\pi \times 1 + 90^\circ}{3} \right) + i \sin \left(\frac{2\pi \times 1 + 90^\circ}{3} \right) \right]$$

$$= 2 \left[-\frac{\sqrt{3}}{2} + i \frac{1}{2} \right]$$

$$= -\sqrt{3} + i$$

When $k=2$,

$$z_3 = 8^{1/3} \left[\cos \left(\frac{2\pi \times 2 + 90^\circ}{3} \right) + i \sin \left(\frac{2\pi \times 2 + 90^\circ}{3} \right) \right]$$

$$= 2 [0 + i(-1)]$$

$$= -2i$$

Hence, cube roots are $\sqrt{3} + i, -\sqrt{3} + i, -2i$. //

20. (C) Ans: Yes, linear permutations and circular permutations are interrelated.

Linear permutation of n objects $= n!$

Circular permutation of n objects $= (n-1)!$

Now, linear permutation $= n! = n \times (n-1)!$

$= n \times$ circular permutation

for eg: If 4 people to be arranged in 4 seats,

No. of linear permutation $= 4!$

If 4 people to be arranged in ^{circular} table,

No. of circular permutation $= 3!$

Now,

$\therefore$ No. of permutation in row $= 4! = 4 \times 3!$

$= 4 \times$ circular

permutation in
table //

Q.N. 21. (a) Solution: Given,

$$\frac{x^2}{9} + \frac{y^2}{16} = 1$$

Differentiating on both sides w.r.t. 'x',

$$\frac{1}{9} \frac{dx^2}{dx} + \frac{1}{16} \frac{dy^2}{dx} = 0$$

$$\text{or, } \frac{1}{9} \cdot 2x + \frac{1}{16} \cdot 2y \frac{dy}{dx} = 0$$

$$\text{or, } \frac{y}{8} \frac{dy}{dx} = -\frac{2x}{9}$$

$$\therefore \frac{dy}{dx} = -\frac{16x}{9y} \Rightarrow \text{slope of tangents}$$

Since the tangents are parallel to x-axis,

$$\frac{dy}{dx} = 0 \Rightarrow -\frac{16x}{9y} = 0$$

$$\therefore x = 0$$

Also,

$$\frac{x^2}{9} + \frac{y^2}{16} = 1$$

$$\text{or, } y^2 = 16$$

$$\therefore y = \pm 4$$

Hence, points where tangents are parallel to x-axis are $(0, 4)$ and $(0, -4)$. //

21. (b) Ans: let 'v' be volume of sphere and 'D' be its diameter.

$$\text{Volume of sphere (v)} = \frac{4}{3} \pi r^3$$

$$\text{or, } v = \frac{4}{3} \pi \left(\frac{D}{2}\right)^3 \quad \left(\because r = \frac{D}{2}\right)$$

$$\therefore v = \frac{\pi D^3}{6}$$

Differentiating on both sides w.r.t. 't',

$$\frac{dv}{dt} = \frac{d}{dD} \left(\frac{\pi D^3}{6} \right)$$

$$\text{or, } \frac{dv}{dt} = \frac{\pi}{6} \cdot 3D^2 \cdot \frac{dD}{dt}$$

$$\text{or, } \frac{dv}{dt} = \frac{\pi D^2}{2} \cdot \frac{dD}{dt}$$

Here, $\frac{dv}{dt}$ is proportional to D^2 , provided

that rate of change of diameter $\left(\frac{dD}{dt}\right)$ is constant. //

21. (c) solution: $\int (\cos x - \sqrt{3} \sin x) dx$

$$= \frac{1}{2} \int \frac{1}{\frac{1}{2} \cos x - \frac{\sqrt{3}}{2} \sin x} dx$$

$$= \frac{1}{2} \int \frac{1}{\sin\left(\frac{\pi}{6}\right) \cos x - \cos\left(\frac{\pi}{6}\right) \sin x} dx$$

$$= \frac{1}{2} \int \frac{1}{\sin\left(\frac{\pi}{6} - x\right)} dx$$

$$= \frac{1}{2} \int \operatorname{cosec}\left(\frac{\pi}{6} - x\right) dx$$

Put $t = \frac{\pi}{6} - x$

Now, $dt = -dx$

$$= -\frac{1}{2} \int \operatorname{cosec} t dt$$

$$= -\frac{1}{2} \ln \left| \tan \frac{t}{2} \right| + C$$

$$= -\frac{1}{2} \ln \left| \tan \left(\frac{\frac{\pi}{6} - x}{2} \right) \right| + C$$

$$= -\frac{1}{2} \ln \left| \tan \left(\frac{\pi}{12} - \frac{x}{2} \right) \right| + C //$$

Q.N. 22. (a) solution: Given,

$$\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$$

or, $\frac{1}{1+y^2} dy = \frac{1}{1+x^2} dx$

Integrating on both sides,

$$\int \frac{1}{1+y^2} dy = \int \frac{1}{1+x^2} dx$$

$$\therefore \tan^{-1} y = \tan^{-1} x + C //$$

22. (b) Ans: The method of partial fraction decomposition is applicable for evaluating an integral of a proper rational fraction i.e. $f(x) = \frac{p(x)}{q(x)}$, where $p(x), q(x)$ are

polynomials and $q(x) \neq 0$. Also, degree of $q(x) >$ degree of $p(x)$. $q(x)$ form should be one of following or all of following $\rightarrow$

- (i) distinct linear factors
- (ii) linear repeating factors
- (iii) Irreducible quadratic factors.

For eg: $I = \int \frac{x^2}{(x+1)^2(x+2)} dx$

Then, $\frac{x^2}{(x+1)^2(x+2)} = \frac{A}{(x+1)^2} + \frac{B}{x+1} + \frac{C}{x+2}$

Or, $x^2 = A(x+2) + B(x+1)(x+2) + C(x+1)^2$

Putting $x = -1, -2$, we get

$A = 1, C = 4$

Also, putting $x = 0$,

$0 = 2A + 2B + C$

Or, $0 = 2 \times 1 + 2B + 4$

$\therefore B = -3$

Now,

$I = \int \frac{dx}{(x+1)^2} - 3 \int \frac{dx}{x+1} + 4 \int \frac{dx}{x+2}$

$= \frac{-1}{x+1} - 3 \ln|x+1| + 4 \ln|x+2| + C //$

22. (c) Solution: Given curve,

$y = x^2$

$\therefore \frac{dy}{dx} = 2x$

$$\frac{dy}{dx} \Big|_{(1,1)} = 2 \times 1 = 2 = m_1$$

Also, $x = y^2$

$$\frac{dx}{dx} = \frac{dy^2}{dx}$$

$$\text{Or, } 1 = 2y \frac{dy}{dx}$$

$$\therefore \frac{dy}{dx} = \frac{1}{2y}$$

$$\frac{dy}{dx} \Big|_{(1,1)} = \frac{1}{2 \times 1} = \frac{1}{2} = m_2$$

Let ' θ ' be angle between the curves.

$$\tan \theta = \pm \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$\text{Or, } \tan \theta = \pm \frac{2 - \frac{1}{2}}{1 + 2 \cdot \frac{1}{2}}$$

$$\text{Or, } \tan \theta = \pm \frac{3}{4}$$

$$\text{Or, } \theta = \tan^{-1} \left(\pm \frac{3}{4} \right)$$

$$\therefore \theta = \tan^{-1} \left(\pm \frac{3}{4} \right) //$$

$$\text{For acute, } \theta = \tan^{-1} \left(\frac{3}{4} \right).$$

" The END "

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Class: XII 'Science'

F.M: 75

Time: 3hr.

Subject: Mathematics (0081)

P.M: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - 'B' [8×5=40]

Short Answer Questions.

12. a) ✓ What is the total number of **distinct permutations** of a set of n objects, if r of them are **of the same kind** and the rest are distinct?
b) ✓ What is the relationship between $P(n, n)$ and $P(n, n - 1)$?
c) State addition principle of counting.
d) What is the exponential form of complex number $r(\cos \theta^\circ + i \sin \theta^\circ)$?
e) ✓ Write the necessary condition for having solution of the system of linear equations by Cramer's Rule.
[1+1+1+1+1=5]
13. a) ✓ How many words can be formed using all the letters of the word **EQUATION** so that
i) all the vowels are together?
ii) consonants occupy odd places? [1+1]
b) ✓ How many even numbers of 3 different digits can be formed from the digits 0, 2, 3, 5, 6, 8? [3]
14. Solve the following system of linear equations by Gauss elimination method.
 $2x + y + 4z - 12 = 0, 8x - 3y + 2z - 20 = 0, 4x + 11y - z - 33 = 0.$
[5]
15. a) ✓ Five boys and five girls sit alternately around round table. In how many ways can this be done? [2]

- b) Using Determinant method, solve the following system of linear equations.

$$2x + y - z = 9$$

$$3x - y + 2z = -1$$

$$4x + y - 3z = 17$$

[3]

16. a) Write the integral of $\int \operatorname{cosech} x \, dx$.

[1]

- b) If $\lim_{x \rightarrow 1} f(x) = 0$, $\lim_{x \rightarrow 1} g(x) = 0$ and $g'(1) \neq 0$, what is the relation

between $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)}$ and $\lim_{x \rightarrow 1} \frac{f'(x)}{g'(x)}$?

[1]

- c) What is the relationship between derivative and antiderivative of function?

[1]

- d) What is the equation of tangent to the curve $y = g(x)$ at the point (h, k) that is parallel to Y -axis?

[1]

- e) What is the derivative of $\tanh^{-1} u$ with respect to u ?

[1]

17. a) If n is natural number and multiple of 3, then prove that

$$1 + \left(\frac{-1+i\sqrt{3}}{2}\right)^n + \left(\frac{-1+i\sqrt{3}}{2}\right)^{2n} = 3$$

[2]

- b) Water flows into an inverted conical tank at the rate of $36 \text{ cm}^3/\text{min}$. When the depth of water is 12 cm, how fast is the level rising if the radius of base and height of the tank is 21 cm and 35 cm respectively.

[3]

18. a) If radius of circle is increasing from 7 cm to 7.1 cm, determine the approximate increase in area.

[2]

- b) Evaluate: $\int \frac{t^2}{t^4 - 2t^2 - 15} dt$.

[3]

19. a) Find $\frac{dy}{dx}$ if $e^{\sinh^{-1} x} = y$

[2]

- b) If $\sqrt{1-x^2} dy + \sqrt{1-y^2} dx = 0$, show that

$$x\sqrt{1-y^2} + y\sqrt{1-x^2} = C \text{ where } C \text{ is constant.}$$

[3]

GROUP - 'C' [3x8=24]

20. a) Evaluate: $\frac{(\sin \frac{\pi}{8} - i \cos \frac{\pi}{8})^8}{(\sin \frac{\pi}{8} + i \cos \frac{\pi}{8})^8}$. [2]
- b) Use De-moivre's theorem to find cube roots of $8i$. [3]
- c) Are linear permutations and circular permutations interrelated?
Justify your answer with a contextual example. *NO* [3]
21. a) Find the points on the curve $\frac{x^2}{9} + \frac{y^2}{16} = 1$ at which the tangents are parallel to x - axis. [2]
- b) The rate of change of volume of sphere is proportional to square of its diameter. Explain [3]
- c) Evaluate: $\int \frac{1}{\cos x - \sqrt{3} \sin x} dx$. [3]
22. a) Solve: $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$. [2]
- b) State the necessary conditions under which the method of partial fraction decomposition is applicable for evaluating an integral.
Justify your answer with suitable example. [3]
- c) Find the angle between the curves $y = x^2$ and $x = y^2$ at the point $(1,1)$. [3]

Best of Luck

FIRST TERM EXAMINATION 2082

Class: XII 'Science'

SET-A

Time: 25 mins.

Subject: Com. Mathematics (0081)

Write the best option among the four options on your answer sheet

GROUP-A (MCQ) [11×1 = 11]

Re-write the correct option of each question in your answer sheet.

1. If $\frac{(n+2)!}{(n-1)!} = 720$, what is the value of n ?
A) 8 B) 9 C) 10 D) 11

2. What is the value of $\frac{(5000)!}{(2500)!}$?
A) $\frac{2^{2500}[1 \times 3 \times 5 \times \dots \times 5000]}{(2500)!}$ B) $2^{2500}[1 \times 3 \times 5 \times \dots \times 4999]$
C) $\frac{2^{2500}[1 \times 3 \times 5 \times \dots \times 4999]}{(2500)!}$ D) $2^{2500}[1 \times 3 \times 5 \times \dots \times 499]$

3. If $z = \sqrt{3} - 1$, then what is the $\arg(\bar{z})$?
A) 90° B) 180° C) 270° D) 0°

4. Which of the following systems of linear equations is *consistent and independent*?
A) $2x - 3y = 5; 4x - 6y = 10$ B) $2x - 3y = 5; 4x - 6y = 5$
C) $2x - 3y = 5; x + 6y = 5$ D) $x - 2y = 5; 3x - 6y = 10$

5. What is the surface area of a sphere if its volume is increasing at the same rate as its radius?
A) 1 unit B) $\frac{1}{2\sqrt{\pi}}$ units C) 4π units D) $\frac{4\pi}{3}$ units

6. For which value of y is tangent to the curve $y = x + \frac{4}{x^2}$ is parallel to the x -axis?
A) $y = 2$ B) $y = 3$ C) $y = 0$ D) $y = 1$